12TH BIOLOGY · CHAPTER 23 · SOLVED EXERCISE

Pharmacology

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. The first step in drugs discovery is _____.

(a) Lead optimization (b) Preclinical Testing
(c) Target identification (d) Hit identification

Answer: (c) Target identification — Scientists first identify the molecule (protein or enzyme) that is involved in causing the disease.

2. Testing in a larger group about 100–300 patients to test the effectiveness and side effects of drugs indicate _____

(a) Phase 1 (b) Phase 2
(c) Phase 3 (d) Phase 4

Answer: (b) Phase 2 — Phase 1 uses 20–100 healthy volunteers and Phase 3 uses 1,000–3,000 patients.

3. Penicillin is highly effective against _____

(a) Gram-positive bacteria (b) Gram-negative bacteria
(c) Streptomyces (d) Penicillium notatum

Answer: (a) Gram-positive bacteria — It is highly effective against gram-positive bacteria, including streptococcal and staphylococcal infections.

4. The _____ antibiotic derived from soil-dwelling bacteria of the genus Streptomyces,

(a) Penicillin (b) Fluoroquinolones
(c) Sulfonamides (d) Tetracycline

Answer: (d) Tetracycline — Penicillin comes from *Penicillium* fungi; the other two are synthetic.

5. The _____drugs are synthetic compounds derived from sulfanilamide,

(a) Penicillin (b) Fluoroquinolones
(c) Sulfonamides (d) Tetracycline

Answer: (c) Sulfonamides — Sulfonamides were discovered by Gerhard Domagk in 1932.

6. Monoclonal antibodies are considered highly affect because

(a) have broad spectrum action (b) are highly specific to antigens
(c) do not bind to any target (d) are non-biological in nature

Answer: (b) are highly specific to antigens — They bind to a single target with high specificity.

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MCQ 6 is printed as "highly affect"; it means "highly effective". "Hit identification" (MCQ 1) and "Phase 4" (MCQ 2) are not mentioned in the chapter.

SECTION 2: SHORT QUESTIONS

1. Briefly describe Stages of Drug Developmental Process?

1. **Preclinical research:** testing in cell cultures (in vitro) and animals (in vivo) for safety, toxicity and efficacy.
2. **Clinical trials:** testing on humans in Phase 1, Phase 2 and Phase 3.
3. **FDA drug review:** regulatory agencies examine the data and approve or reject the drug.
4. **Manufacturing and marketing:** large-scale production of the approved drug.
5. **Post-marketing surveillance:** continuous monitoring for long-term and rare side effects.

2. Write side effects of Penicillin?

The common side effects of penicillin antibiotics are nausea, vomiting, diarrhoea, abdominal pain and skin rashes.

3. Write a mode of Tetracycline action?

1. **Bacteriostatic effect:** It stops bacteria from growing and multiplying rather than killing them outright.
2. **Gram-negative/positive action:** It is effective against both Gram-positive and Gram-negative bacteria.
3. **Targeting ribosomes:** It enters the bacterial cell and binds reversibly to the 30S ribosomal subunit.
4. **Blocking protein production:** By binding to the 30S ribosome, it prevents aminoacyl-tRNA from binding to the acceptor site. This inhibits translation and protein synthesis.

4. Differentiate between antiviral and antiretroviral drugs?

Antiviral drugs	Antiretroviral drugs
Used to treat viral infections in general.	Designed to treat illnesses caused by retroviruses, including HIV.
Prevent the growth and multiplication of viruses within the body.	Stop HIV from proliferating and aid the immune system's defence.
Act against a wide range of viruses, e.g., hepatitis, influenza and herpes viruses.	Act against retroviruses (viruses with RNA as genetic material that replicate by reverse transcriptase).
All antivirals are not antiretrovirals.	All antiretrovirals are antivirals.

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5. What are the advantages of monoclonal antibodies?

1. Compared with traditional drugs, they offer lower toxicity and the potential for long-lasting effects.
2. They have the least side effects compared with conventional chemotherapeutics.
3. They have high specificity, binding to a single target with minimal variability.
4. They provide targeted therapy in cancer, reducing damage to healthy tissues.
5. They are easy to prepare and are used for research purposes.
6. They reduce the risk of drug interactions.

SECTION 3: LONG QUESTIONS

1. Describe the essential steps for the drug discovery and developmental process?

Drug discovery and drug development produce safe and effective medicines. Before a drug is approved for public use, it is tested in several stages for safety, efficacy and quality.

A. Stages of drug discovery

Drug discovery is the initial stage of finding a new drug for a disease. Its stages are:

1. **Target identification:** Scientists identify a specific molecule, such as a protein or enzyme, that is involved in causing the disease and can be affected by a drug.
2. **Target validation:** Researchers confirm by experiments that the target plays an important role in the disease and that acting on it can produce therapeutic effects.
3. **Lead compound identification:** Thousands of natural or synthetic compounds are screened to find substances that interact effectively with the target. These promising substances are called **lead compounds**.
4. **Lead optimization:** The lead compounds are modified to improve their effectiveness, reduce toxicity and enhance their drug-like properties.

B. Drug developmental process

Drug development is the process of testing and preparing the discovered drug for public use. Its stages are:

1. **Preclinical research:** The compounds are tested *in vitro* (cell-based culture) and *in vivo* (animals) to measure safety, toxicity and efficacy before human trials.
2. **Clinical trials:** The drug is tested on humans in three phases (see the table).
3. **FDA drug review:** Regulatory agencies (FDA or EMA) examine all the data of the clinical trials and approve or reject the drug before it is supplied to the market.
4. **Manufacturing and marketing:** The approved drug is produced on a large scale in pharmaceutical industries and made available for medical use.
5. **Post-marketing surveillance:** After release, the drug is continuously monitored for long-term effects and rare side effects.

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Phase	Who is tested	Purpose
Phase 1	Small group (20–100) of healthy volunteers	To determine safety and dosage
Phase 2	Larger group (100–300) of patients	To test effectiveness and side effects
Phase 3	Large-scale trials (1,000–3,000)	To confirm efficacy and monitor adverse reactions



Drug discovery (first four steps) and development (last five steps)

2. Explain different type of antibiotics. Describe their mode of action.

Antibiotics are drugs that treat bacterial infections by killing bacteria or stopping their growth. They are classified mainly by their mechanism of action and chemical structure. The four major classes are penicillin, tetracycline, fluoroquinolones and sulfonamides.

1. Penicillin

Penicillin is a group of bactericidal antibiotics derived from *Penicillium* fungi. It treats bacterial infections by inhibiting bacterial cell wall synthesis. Alexander Fleming discovered it in 1928. It is highly effective against gram-positive bacteria, including streptococcal and staphylococcal infections. Common side effects: nausea, vomiting, diarrhoea, abdominal pain and skin rashes.

Mode of action

1. **Targeting bacteria:** It targets bacteria only, because animal cells lack the cell wall that the drug attacks.
2. **Cell wall inhibition:** It interrupts the cross-linking of peptidoglycan (murein) chains, so new cell wall is not synthesised.
3. **Lysing the bacterial cell:** The cell wall can no longer maintain its structural integrity, and the cell bursts (lysis). So it kills bacteria rather than just inhibiting growth.
4. **Resistance mechanism:** Some bacteria defend themselves by producing specific enzymes such as beta-lactamases.

2. Tetracycline

Tetracycline is derived from soil-dwelling bacteria of the genus *Streptomyces*. It is a bacteriostatic oral antibiotic used for bacterial infections like chlamydia and syphilis. It inhibits bacterial protein synthesis. It should not be taken with dairy or iron supplements. Common side effects: nausea, vomiting, diarrhoea and sun sensitivity.

Mode of action

1. **Bacteriostatic effect:** It stops bacteria from growing and multiplying rather than killing them outright.
2. **Gram-negative/positive action:** It is effective against both Gram-positive and Gram-negative bacteria.
3. **Targeting ribosomes:** It enters the bacterial cell and binds reversibly to the 30S ribosomal subunit.
4. **Blocking protein production:** By binding to the 30S ribosome, it prevents aminoacyl-tRNA from binding to the acceptor site. This inhibits translation and protein synthesis.

3. Fluoroquinolones

Fluoroquinolones are synthetic antibacterial agents, derived as a by-product of chloroquine synthesis (chloroquine is a synthetic drug made by modifying quinine, a natural alkaloid of the bark of the South American *Cinchona* tree). They were discovered by George Lesher (1962). They are bactericidal antibiotics that inhibit bacterial DNA replication. Rare but significant side effects: tendon ruptures, peripheral neuropathy, drug resistance and cardiac issues.

Mode of action

1. **Bactericidal effect:** They trap the enzymes of DNA replication in a complex that causes double-strand DNA breaks. The bacteria cannot multiply and die.
2. **Targeting topoisomerases:** They enter the bacterial cell and inhibit its type II topoisomerases.
3. **Targeting topoisomerase IV:** mainly in Gram-positive bacteria; this stops the separation of replicated DNA.
4. **Targeting DNA gyrase:** mainly in Gram-negative bacteria; this stops the uncoiling of DNA during replication.

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4. Sulfonamides

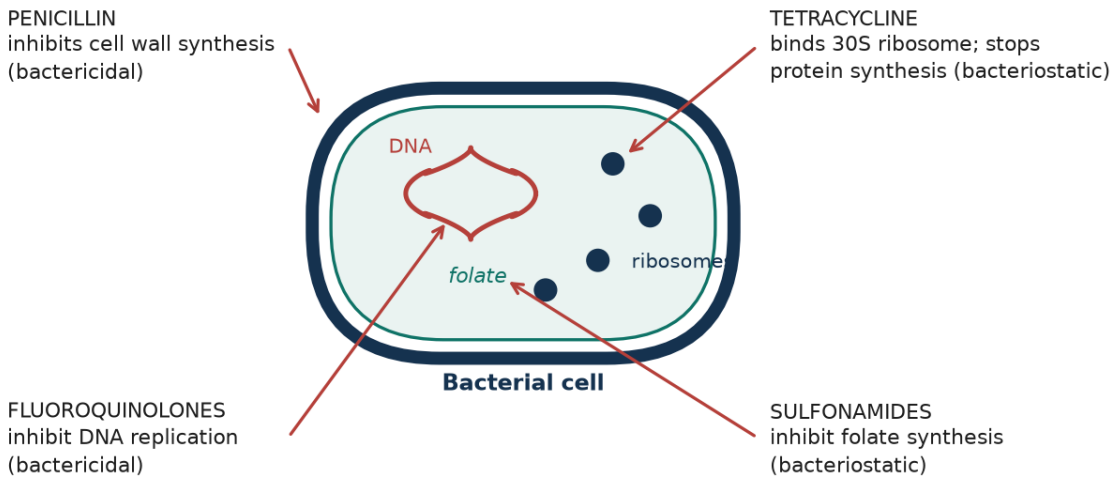
Sulfonamides are synthetic compounds derived from sulfanilamide, discovered by Gerhard Domagk in 1932. They are bacteriostatic antimicrobial drugs that inhibit the bacterial synthesis of folic acid (folate), which is essential for DNA synthesis and bacterial growth. They are mainly used for urinary tract infections and skin infections and are often given orally. Side effects: rashes, nausea, photosensitivity and kidney damage.

Mode of action

1. **Bacteriostatic effect:** They do not kill bacteria immediately but inhibit their growth, so that the body's immune system can destroy them.
2. **Selective toxicity:** Bacteria must build folate from scratch, whereas humans simply absorb it from food. So the drug starves bacteria while leaving human cells untouched.
3. **Competitive inhibition:** They bind to the enzyme that makes folate, taking the place of para-amino benzoic acid (PABA) in the folate pathway.
4. **Combined therapy:** They are often combined with other agents, like trimethoprim, for a stronger bactericidal effect.
5. **Broad spectrum:** They are effective against both Gram-positive and Gram-negative bacteria (urinary tract infections, acne, respiratory infections).

Summary

Class	Source / discovery	Effect	Target in bacteria
Penicillin	*Penicillium* fungi; Alexander Fleming (1928)	Bactericidal	Cell wall synthesis
Tetracycline	Soil bacteria of genus *Streptomyces*	Bacteriostatic	Protein synthesis (30S ribosomal subunit)
Fluoroquinolones	Synthetic; George Lesher (1962)	Bactericidal	DNA replication (type II topoisomerases)
Sulfonamides	Synthetic, from sulfanilamide; Gerhard Domagk (1932)	Bacteriostatic	Folic acid (folate) synthesis



Targets of the four classes of antibiotics in a bacterial cell

3. Describe the Monoclonal Antibodies and their advantages?

Monoclonal antibodies are artificially laboratory-produced molecules that enhance the immune system's ability to fight against pathogens. They target specific antigens, which makes them effective for treating cancer, autoimmune disorders and infectious diseases. They are integral to personalized medicine, because treatment can be tailored to the profile of the individual patient.

Advantages

1. Compared with traditional drugs, they offer lower toxicity and the potential for long-lasting effects.
2. They have the least side effects compared with conventional chemotherapeutics.
3. They have high specificity, binding to a single target with minimal variability.
4. They provide targeted therapy in cancer, reducing damage to healthy tissues.
5. They are easy to prepare and are used for research purposes.
6. They reduce the risk of drug interactions.

New technologies allow thousands of potential antibodies to be tested at the same time. This shortens development time and allows faster responses to urgent health challenges such as infectious diseases and cancer.

INQUISITIVE QUESTIONS

1. Predict the challenges the scientists may face during drug development process and propose possible solutions.

Challenge	Possible solution
Finding the right target: the chosen molecule may not really control the disease.	Careful target validation by experiments before going further.
Thousands of compounds must be screened to find a few lead compounds.	Fast, automated screening and computer-aided selection of compounds.
Toxicity: a compound that works may also harm the body.	Lead optimization to reduce toxicity; thorough preclinical tests in cells and animals.
Results in animals may not be repeated in humans.	Step-by-step clinical trials, starting with a small group in Phase 1.
Rare or long-term side effects do not appear in trials.	Large Phase 3 trials and post-marketing surveillance.
The process takes many years and is very costly; the drug may still be rejected by the regulator.	Good study design, cooperation of scientists, researchers and regulatory authorities, and complete, honest data.
Bacteria and viruses develop resistance to the new drug.	Combined therapy and continued research for new drugs.

The textbook describes the stages only; the challenges and solutions are reasoned from those stages.

2. Assess the possible consequences of the misuse and overuse of antibiotics on human health.

- **Antibiotic resistance:** bacteria that survive the drug multiply and pass on their resistance (e.g., bacteria producing beta-lactamases against penicillin). The antibiotic then stops working, and infections become difficult to treat.
- **Longer and costlier illness:** resistant infections need stronger drugs, longer hospital stays and may cause death.
- **More side effects:** unnecessary use exposes the patient to nausea, vomiting, diarrhoea, rashes, photosensitivity, kidney damage or tendon rupture, without any benefit.
- **Loss of useful bacteria:** antibiotics also kill the helpful bacteria of the gut, which upsets digestion and lets harmful microbes grow.
- **No effect on viral diseases:** antibiotics do not act on viruses, so taking them for flu or the common cold only adds to resistance.

So antibiotics should be taken only on a doctor’s prescription, in the correct dose and for the complete course.

Only the resistance mechanism of bacteria and the side effects are given in the textbook; the other points are general knowledge.